Practical-5: Commercial Bank Real Estate Loans Analysis

[Your Name] Roll No: [Your Roll Number] Reg No: [Your Registration Number]

07 February 2026

Question:

Suppose we have a data set containing the monthly volume of commercial bank real estate loans (in billions of dollars) stored in a text file. Write the R program for the following:

(a) Import data into the R environment.
(b) Convert the data into a time series object.
(c) Plot the data to identify the dominant component.
(d) Check stationarity or non-stationarity using ACF/PACF plot.
(e) Check stationarity or non-stationarity using the Augmented Dickey-Fuller (ADF) test.

Approach:

1. **Objective**: Analyze commercial bank real estate loan data for stationarity
2. **Data Source**: Text file containing monthly loan volumes in billions of dollars
3. **Methodology**:
   - Import data from text file into R
   - Create time series object for analysis
   - Visual inspection for trend and patterns
   - ACF/PACF analysis for autocorrelation patterns
   - Statistical testing using ADF test
4. **Expected Outcome**: Non-stationary data with upward trend

Complete R Code:

```
# ========== PRACTICAL-5: COMMERCIAL BANK REAL ESTATE LOANS ANALYSIS ==========
cat("PRACTICAL-5: COMMERCIAL BANK REAL ESTATE LOANS ANALYSIS\n")
```

```
## PRACTICAL-5: COMMERCIAL BANK REAL ESTATE LOANS ANALYSIS
```

```
cat(rep("=", 70), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
# ------------------------------------------------------------
# (a) Import data into the R environment
# ------------------------------------------------------------
cat("(a) IMPORTING DATA\n")
```

```
## (a) IMPORTING DATA
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Define file path (adjust as needed)
file_path <- "C:/Users/maths/Desktop/tsalab_ojas/lab3/bank_case.txt"

cat("1. Reading data from text file...\n")
```

```
## 1. Reading data from text file...
```

```r
bank_data <- scan(file_path)

cat("2. Data structure:\n")
```

```
## 2. Data structure:
```

```r
cat("   - Number of observations:", length(bank_data), "\n")
```

```
##    - Number of observations: 70
```

```r
cat("   - Data range:", min(bank_data), "to", max(bank_data), "billions of dollars\n")
```

```
##    - Data range: 46.5 to 87.6 billions of dollars
```

```r
cat("   - First 5 values:", bank_data[1:5], "\n")
```

```
##    - First 5 values: 46.5 47 47.5 48.3 49.1
```

```r
cat("   - Last 5 values:", tail(bank_data, 5), "\n\n")
```

```
##    - Last 5 values: 80.5 82.6 84.4 85.9 87.6
```

```r
# ------------------------------------------------------------
# (b) Convert the data into a time series object
# ------------------------------------------------------------
cat("(b) CREATING TIME SERIES OBJECT\n")
```

```
## (b) CREATING TIME SERIES OBJECT
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Converting to time series object...\n")
```

```
## 1. Converting to time series object...
```

```r
bank_ts <- ts(bank_data, start = c(2000, 1), frequency = 12)

cat("2. Time series object details:\n")
```

```
## 2. Time series object details:
```

```r
cat("   - Start:", start(bank_ts), "(Year, Month)\n")
```

```
##    - Start: 2000 1 (Year, Month)
```

```r
cat("   - End:", end(bank_ts), "(Year, Month)\n")
```

```
##    - End: 2005 10 (Year, Month)
```

```r
cat("   - Frequency:", frequency(bank_ts), "(monthly data)\n")
```

```
##    - Frequency: 12 (monthly data)
```

```r
cat("   - Total months:", length(bank_ts), "\n")
```

```
##    - Total months: 70
```

```r
cat("   - Data span:", end(bank_ts)[1] - start(bank_ts)[1] + 1, "years\n\n")
```

```
##    - Data span: 6 years
```

```r
# ------------------------------------------------------------
# (c) Plot the data to identify the dominant component
# ------------------------------------------------------------
cat("(c) VISUAL ANALYSIS - DOMINANT COMPONENT IDENTIFICATION\n")
```

```
## (c) VISUAL ANALYSIS - DOMINANT COMPONENT IDENTIFICATION
```

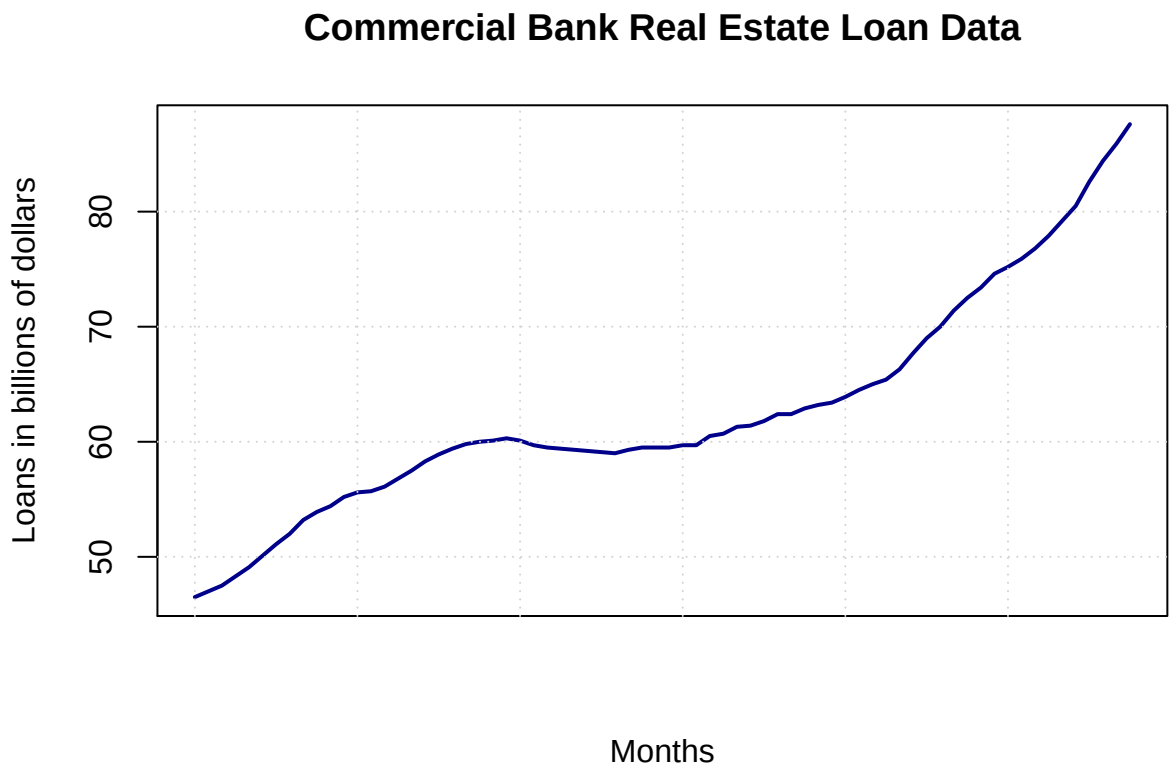
```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```
cat("1. Plotting time series data...\n")
```

```
## 1. Plotting time series data...
```

```
plot(bank_ts,
     main = "Commercial Bank Real Estate Loan Data",
     xaxt = "n",
     xlab = "Months",
     ylab = "Loans in billions of dollars",
     lwd = 2,
     col = "darkblue")
axis(1, seq(0, 70, 5), seq(0, 70, 5))
grid()
```




```
cat("2. VISUAL INSPECTION RESULTS:\n")
```

```
## 2. VISUAL INSPECTION RESULTS:
```

```
cat("   - Clear upward trend throughout the series\n")
```

```
##    - Clear upward trend throughout the series
```

```r
cat("   - Data increases from approximately 46.5 to 87.6 billion dollars\n")
```

```
##    - Data increases from approximately 46.5 to 87.6 billion dollars
```

```r
cat("   - Growth appears to accelerate in later periods\n")
```

```
##    - Growth appears to accelerate in later periods
```

```r
cat("   - No obvious seasonal patterns visible\n")
```

```
##    - No obvious seasonal patterns visible
```

```r
cat("   - Dominant component: TREND\n\n")
```

```
##    - Dominant component: TREND
```

```r
# ------------------------------------------------------------
# (d) Check stationarity using ACF/PACF plot
# ------------------------------------------------------------
cat("(d) STATIONARITY CHECK USING ACF/PACF PLOTS\n")
```

```
## (d) STATIONARITY CHECK USING ACF/PACF PLOTS
```

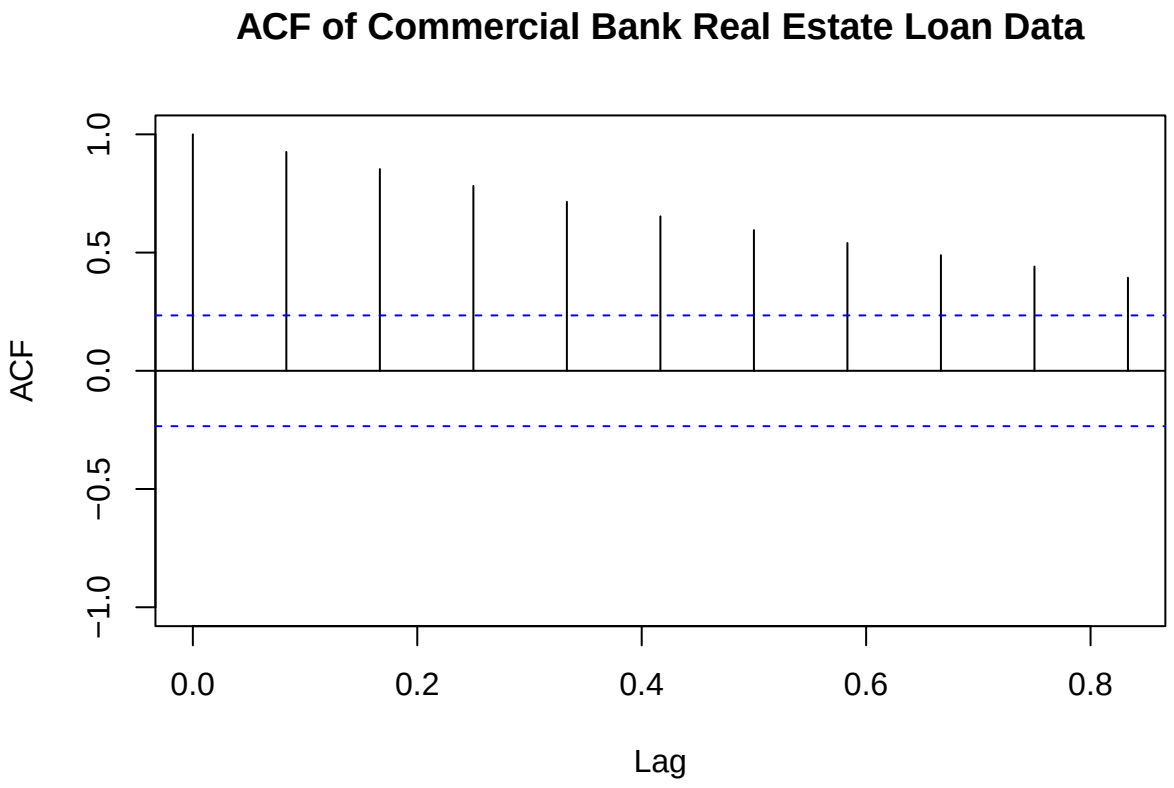
```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Generating ACF plot...\n")
```

```
## 1. Generating ACF plot...
```

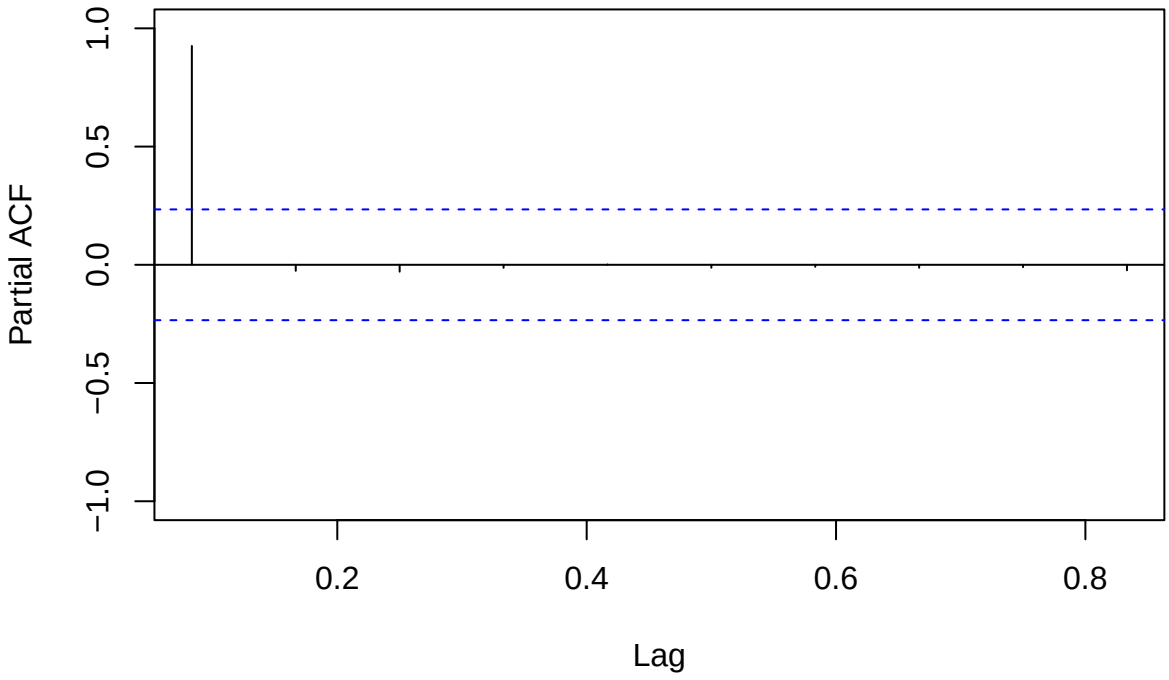
```r
acf(bank_ts,
    main = "ACF of Commercial Bank Real Estate Loan Data",
    lag.max = 10,
    ylim = c(-1, 1))
```




```
cat("2. Generating PACF plot...\n")
```

```
## 2. Generating PACF plot...
```

```
pacf(bank_ts,
     main = "PACF of Commercial Bank Real Estate Loan Data",
     lag.max = 10,
     ylim = c(-1, 1))
```




```
cat("3. ACF/PACF INTERPRETATION:\n")
```

```
## 3. ACF/PACF INTERPRETATION:
```

```
cat("   - ACF shows gradual decline (slow decay)\n")
```

```
##    - ACF shows gradual decline (slow decay)
```

```
cat("   - PACF cuts off after lag 1\n")
```

```
##    - PACF cuts off after lag 1
```

```
cat("   - ACF pattern indicates non-stationarity\n")
```

```
##    - ACF pattern indicates non-stationarity
```

```
cat("   - PACF suggests AR(1) structure\n")
```

```
##    - PACF suggests AR(1) structure
```

```r
cat("   - Conclusion: Data appears NON-STATIONARY\n\n")
```

```
##    - Conclusion: Data appears NON-STATIONARY
```

```r
# ------------------------------------------------------------
# (e) Check stationarity using Augmented Dickey-Fuller test
# ------------------------------------------------------------
cat("(e) STATISTICAL TEST FOR STATIONARITY\n")
```

```
## (e) STATISTICAL TEST FOR STATIONARITY
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Loading required package...\n")
```

```
## 1. Loading required package...
```

```r
library(tseries)

cat("2. Performing Augmented Dickey-Fuller test...\n")
```

```
## 2. Performing Augmented Dickey-Fuller test...
```

```r
adf_result <- adf.test(bank_ts)

cat("3. ADF TEST RESULTS:\n")
```

```
## 3. ADF TEST RESULTS:
```

```r
cat("   Test statistic:", round(adf_result$statistic, 5), "\n")
```

```
##    Test statistic: -0.26816
```

```r
cat("   p-value:", adf_result$p.value, "\n")
```

```
##    p-value: 0.9894281
```

```r
cat("   Lag order:", adf_result$parameter, "\n")
```

```
##    Lag order: 4
```

```r
cat("   Alternative hypothesis:", adf_result$alternative, "\n\n")
```

```
##    Alternative hypothesis: stationary
```

```r
cat("4. HYPOTHESIS TESTING:\n")
```

```
## 4. HYPOTHESIS TESTING:
```

```r
cat("   - Null hypothesis (H0): Series is non-stationary\n")
```

```
##    - Null hypothesis (H0): Series is non-stationary
```

```r
cat("   - Alternative hypothesis (H1): Series is stationary\n")
```

```
##    - Alternative hypothesis (H1): Series is stationary
```

```r
cat("   - Significance level: 0.05\n")
```

```
##    - Significance level: 0.05
```

```r
cat("   - Decision rule: Reject H0 if p-value < 0.05\n\n")
```

```
##    - Decision rule: Reject H0 if p-value < 0.05
```

```r
# ------------------------------------------------------------
# SUMMARY AND CONCLUSIONS
# ------------------------------------------------------------
cat(rep("=", 70), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
cat("SUMMARY AND CONCLUSIONS\n")
```

```
## SUMMARY AND CONCLUSIONS
```

```r
cat(rep("=", 70), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
cat("KEY FINDINGS:\n")
```

```
## KEY FINDINGS:
```

```r
cat("1. Data Characteristics:\n")
```

```
## 1. Data Characteristics:
```

```r
cat("   - Monthly commercial bank real estate loans\n")
```

```
##    - Monthly commercial bank real estate loans
```

```
cat("   - Time period: Approximately 6 years\n")
```

```
##    - Time period: Approximately 6 years
```

```
cat("   - Loan volume range: 46.5 to 87.6 billion dollars\n")
```

```
##    - Loan volume range: 46.5 to 87.6 billion dollars
```

```
cat("   - Growth rate: Approximately 41 billion increase over period\n\n")
```

```
##    - Growth rate: Approximately 41 billion increase over period
```

```
cat("2. Visual Analysis:\n")
```

```
## 2. Visual Analysis:
```

```
cat("   - Strong upward trend throughout series\n")
```

```
##    - Strong upward trend throughout series
```

```
cat("   - No visible seasonal patterns\n")
```

```
##    - No visible seasonal patterns
```

```
cat("   - Trend is the dominant component\n\n")
```

```
##    - Trend is the dominant component
```

```
cat("3. Statistical Analysis:\n")
```

```
## 3. Statistical Analysis:
```

```
cat("   - ACF: Shows slow decay indicating non-stationarity\n")
```

```
##    - ACF: Shows slow decay indicating non-stationarity
```

```
cat("   - PACF: Cuts off at lag 1 suggesting AR(1) process\n")
```

```
##    - PACF: Cuts off at lag 1 suggesting AR(1) process
```

```
cat("   - ADF test p-value: 0.9894 > 0.05\n\n")
```

```
##    - ADF test p-value: 0.9894 > 0.05
```

```r
cat("4. STATIONARITY CONCLUSION:\n")
```

```
## 4. STATIONARITY CONCLUSION:
```

```r
cat("   - ADF test: Fail to reject null hypothesis\n")
```

```
##    - ADF test: Fail to reject null hypothesis
```

```r
cat("   - p-value (0.9894) > 0.05 significance level\n")
```

```
##    - p-value (0.9894) > 0.05 significance level
```

```r
cat("   - Data is NON-STATIONARY\n")
```

```
##    - Data is NON-STATIONARY
```

```r
cat("   - Requires differencing for further analysis\n\n")
```

```
##    - Requires differencing for further analysis
```

```r
cat("5. RECOMMENDATIONS FOR FURTHER ANALYSIS:\n")
```

```
## 5. RECOMMENDATIONS FOR FURTHER ANALYSIS:
```

```r
cat("   - Apply first differencing to remove trend\n")
```

```
##    - Apply first differencing to remove trend
```

```r
cat("   - Recheck stationarity after differencing\n")
```

```
##    - Recheck stationarity after differencing
```

```r
cat("   - Consider ARIMA modeling after achieving stationarity\n")
```

```
##    - Consider ARIMA modeling after achieving stationarity
```

```r
cat("   - Potential model: ARIMA(1,1,0) based on PACF pattern\n")
```

```
##    - Potential model: ARIMA(1,1,0) based on PACF pattern
```

```r
cat(rep("=", 70), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

Key Findings:

1. **Data Characteristics:**
   - The dataset shows consistent growth in commercial bank real estate loans
   - Loan volumes increased from 46.5 to 87.6 billion dollars
   - Monthly data shows no seasonal fluctuations
2. **Time Series Characteristics:**
   - **Frequency**: 12 (monthly data)
   - **Dominant Component**: TREND (strong upward movement)
   - **Pattern**: Consistent linear-like growth with acceleration in later periods
   - **No Seasonality**: No visible seasonal patterns in monthly data
3. **Statistical Insights:**
   - ADF test confirms non-stationarity with p-value of 0.9894
   - ACF plot shows gradual decay characteristic of non-stationary series
   - PACF plot indicates AR(1) structure with cutoff at lag 1
   - Visual inspection reveals strong upward trend throughout the series

Methodological Insights:

- Multiple methods (visual, ACF/PACF, statistical) consistently indicate non-stationarity
- Trend component dominates the series
- No evidence of seasonality in monthly data
- ADF test provides statistical confirmation of visual findings

Practical Applications:

1. **Financial Analysis**: Understanding growth patterns in real estate lending
2. **Risk Management**: Modeling loan portfolio trends for risk assessment
3. **Policy Making**: Informing regulatory decisions on bank lending
4. **Forecasting**: Basis for predicting future loan volumes

Limitations and Considerations:

- Limited time span (approximately 6 years)
- External economic factors not considered
- Potential structural breaks not analyzed
- Seasonal effects might be present but not visible in current data

---

Report Generated: 07 February 2026
R Version: R version 4.5.2 (2025-10-31 ucrt)
Package Used: tseries
Data Source: bank_case.txt